

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2017



Hum 1217

(Principles of Economics)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is scarcity and choice? Explain production possibilities frontier on the basis of choice, scarcity and opportunity cost. (15)
- (b) Mention the key problems of an economic organization. (10)
- (c) How do the basic problems be solved through various economic systems? (10)
2. (a) Define law of demand. Explain the determinants of demand. (15)
- (b) Solve the problem: (20)
There are 10,000 identical individuals in the market for commodity X, each with a demand function is given by $Q_{dx}=12-2P_x$, and 1,000 identical producers of commodity X, each with a supply function is given by $Q_{sx}=20P_x$.
(i) Obtain equilibrium price and quantity.
In that case, if the government decides to collect a sales tax of \$2 per unit sold from each of the 1,000 identical sellers of commodity X-
(ii) What effect does this have on the equilibrium price and quantity?
(iii) Who actually pays the tax?
(iv) What is the amount of taxes collected by the government?
3. (a) What is production function? Is land capital? (10)
- (b) Explain the law of diminishing returns. (15)
- (c) What are the main factors of production? Explain. (10)
4. (a) Define cost. Explain Total cost = Fixed cost + Variable cost. (10)
- (b) What is short-run? Describe short-run equilibrium of a firm under perfect competition. (15)
- (c) Explain the shut-down condition of a perfect competitive firm. (10)

Section – B

5. (a) Explain the idea of Real GDP and Nominal GDP. List the four components of GDP. Give an example of each. (15)
- (b) In what way is a university degree a form of capital? (10)
- (c) Explain how higher saving leads to a higher standard of living. What might defer a policy maker from trying to raise the rate of savings? (10)
6. (a) Explain the idea of inflation, demand pull inflation and cost push inflation. (15)
- (b) What is national, private and public savings? Describe a change in tax-code that might increase private savings. If this policy was implemented, how would it affect the market for loanable fund? (20)
7. (a) What is the role of the financial system? Name and describe two markets that are the part of the financial system in our economy. (15)
- (b) Suppose GDP is \$8 trillion, taxes are \$1.5 trillion, private savings is \$0.5 trillion and the public savings is \$0.2 trillion. Consider this economy is closed. Calculate consumption, government purchase, national savings and investment and explain. (20)
8. (a) Define bank. Discuss the role of banks in economic development of Bangladesh. (15)
- (b) Discuss the Functions of commercial bank and central bank of a country separately. (20)

Khulna University of Engineering & Technology
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BURP 1st Year 2nd Term Regular Examination, 2017
Math 1217
(Mathematics - II)



Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. Calculate:

(a) $\int \frac{dx}{(x-1)\sqrt{(2x^2-8x-1)}}$ (13)

(b) $\int \frac{e^x}{x} (1+x \log x) dx$ (10)

(c) $\int \frac{dx}{5+4 \sin 2x}$ (12)

2. (a) Calculate:

$\int \frac{dx}{x^3-1}$ (10)

(b) Obtain a reduction formula for $\int \cos^n x dx$ (12)

(c) If $u_n = \int_0^1 x^n \tan^{-1} x dx$, then prove that $(n+1)u_n + (n-1)u_{n-2} = \frac{\pi}{2} - \frac{1}{n}$ (13)

3. Evaluate:

(a) $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$ (11)

(b) $\int_0^{\pi/2} \log \cos x dx$ (13)

(c) $\int_0^1 x^3 (1-x^2)^{5/2} dx$ (11)

4. (a) Evaluate

$\lim_{n \rightarrow \infty} \left[\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{2n} \right]$ (11)

(b) Find the area bounded by the curve $x^{2/3} + y^{2/3} = a^{2/3}$ (12)

(c) Find the area of the segment cut off from the parabola $y^2 = 4x$ by the straight line $y = 2x$. (12)



Section – B

5. (a) Define continuity of a function at a point. A function $f(x)$ is defined as follows: (15)

$$f(x) = \begin{cases} x - 3 & \text{when } x \geq 3 \\ -x + 3 & \text{when } x < 3 \end{cases}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 3$.

- (b) Evaluate (10)

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$$

- (c) If $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$, find $\frac{dy}{dx}$ (10)

6. (a) Show that $f(x) = \sin x (1 + \cos x)$ has a maximum value at $x = \frac{\pi}{3}$. Also find the maximum value. (12)

- (b) If $y = \sin(m \sin^{-1} x)$, then find the value of (13)

$$(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n.$$

- (c) If $f(x, y, z) = x^2y + y^2z + z^2x$, then find the values of (10)

$$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y \partial z}, \frac{\partial^2 f}{\partial z \partial x}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial z^2}$$

7. Solve:

- (a) $\frac{dy}{dx} = \frac{2x - 2y + 3}{x - y + 5}$ (12)

- (b) $\left(y + \frac{y^3}{3} + \frac{x^2}{2} \right) dx + \frac{1}{4}(x + xy^2) dy = 0$ (12)

- (c) $x \frac{dy}{dx} + 2y = \log x$ (11)

8. (a) Solve $\frac{d^2y}{dx^2} + y = \tan x$ by the method of variation of parameters. (13)

- (b) Solve $(D^2 - 2D + 1)y = x^2e^x$ (12)

- (c) Solve $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = e^{-3x} + \cos 2x$. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2017
URP 1211
(Urban Planning Principles)

Full Marks: 210

Time: 3.00 hrs

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define urban planning? Explain the scopes of urban planning with examples. (15)
(b) Write down the differences between satellite town and new town. (08)
(c) Clarify the concept of “Town Centre”. Discuss about the functions of a town centre. Use proper example. (12)
2. (a) “Each Neighbourhood is a distinct physical unit with self sufficiency in local needs” – Justify the statement. (10)
(b) Mention the causes, consequences and remedies of disappearance of open space in the cities and towns of Bangladesh with examples in your own words. (15)
(c) Write down the principles of creating a successful Town Centre. (10)
3. (a) “Proper balance among four objects of Town Planning is important for designing a Town” – Justify the statement. (15)
(b) Discuss the present condition of open space in the cities and towns of Bangladesh. (12)
(c) Write down some criticisms of Residential Developments with proper examples. (08)
4. (a) Describe the Evolvement Process from “Satellite Towns” to New Towns. (15)
(b) Write short notes on the followings (Answer any four) (20)
 - i. Functions of Open Space
 - ii. Greenbelt
 - iii. Components of Town Centre
 - iv. Who are Leisure users and tourists in a Railway Station?
 - v. Draw a diagram to show maximum walking distance of a pedestrian to station facilities from station entrance.

Section – B

5. (a) What is the basic concept of “Central Place Theory”? Discuss about the three principles of “Central Place Theory” with necessary sketch. (20)
- (b) Write about the spatial considerations addressed in case of sitting commercial land use. Use necessary examples. (15)
6. (a) “Eco-industrial parks are designed to minimize the conflict between environment and industrialization process”- Explain the statement in context of Bangladesh. (10)
- (b) What is understood by industrial agglomeration? Mention the advantages and disadvantages of industrial agglomeration in context of Bangladesh. (10)
- (c) What is understood by industrial linkage? Give examples of different types of industrial linkages in context of Bangladesh and sketch necessary diagrams. (15)
7. (a) What is understood by Commercial Area? Discuss about the characteristics of commercial land uses with proper example. (12)
- (b) For the purpose of issuance of Environmental Clearance Certificate, the industrial units and projects are classified into four categories. Give three examples of each category. (08)
- (c) “Industrial park includes all crucial facilities for industrial development” – write about the service facilities available in any industrial park with necessary example. (15)
8. (a) Explain Floor Area Ratio (FAR). “If FAR is same, increasing height of building reduces building coverage and vice versa”-justify this statement with examples. (15)
- (b) Write short notes on any four of the following issues: (20)
- i) Types of towns according to function
 - ii) Linear City
 - iii) Urbanism
 - iv) Means to Control Location of Industry
 - v) Runways based on orientation



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1stYear 2ndTerm Regular Examination, 2017
URP 1251
(Introduction to Architectural History)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is Architectural Heritage? (05)
(b) Shortly describe the factors that influence Architecture and Heritage. (15)
(c) Write short notes on the following topics: (15)
 - i. The Sphinx
 - ii. The Erechtheum
 - iii. The Roman road construction
2. (a) Explain the Greek Architectural characteristics with necessary sketches. (15)
(b) Draw the schematic master plan of Greek Agora. (10)
(c) Describe three major differences between Greek and Roman Architecture. (10)
3. (a) Explain the Architectural elements of Gothic period with necessary sketches. (20)
(b) What Architectural Characteristics Renaissance style emphasized on? (05)
(c) Write short notes on the following topics – (10)
 - i. New York five
 - ii. Organic Architecture
4. (a) Explain the following topics with necessary sketches – (25)
 - i. Cubism
 - ii. Abbot Suger
 - iii. Mastaba
 - iv. Greek Optical Correction
 - v. Roman Composite Wall
(b) What are the causes of failure of Modern Movement in Architecture? (10)

Section – B

5. (a) Discuss "Stupa" with necessary sketches. The Stupa symbolizes five purified elements – Explain. (15)
(b) Describe the philosophical interpretation of Buddhist Vihara with necessary sketches. (10)
(c) Write down the classifications of Hindu temple. (10)
6. (a) Describe the architectural features of Kantoji Temple with necessary sketches. (20)
(b) Mention the major components of the Khan-e-Jahan style. Elaborate the components. (15)
7. (a) "Curzon Hall is a blend of European and Mughal style" - Justify the statement with necessary sketches. (15)
(b) "Bungalow Style evolved from the Bengal architecture" - Justify the statement. (15)
(c) Mention the basic features of a "Bungalow" (05)
8. Write short notes on any 5 (five) of the following: (35)
 - (i) Components of a Bengali Hut
 - (ii) The Timeless Landscape
 - (iii) The Shait Gambuj (Sixty Dome) Mosque
 - (iv) Indo-Islamic Style of Temples
 - (v) Background of Colonial Architecture in Bengal
 - (vi) Later developments of Bungalow
 - (vii) Char Chala

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2017
URP 1281
(Surveying and Cartography)



Full Marks: 210

Time: 3.00 hrs

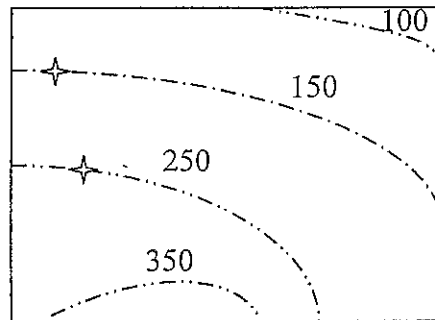
- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

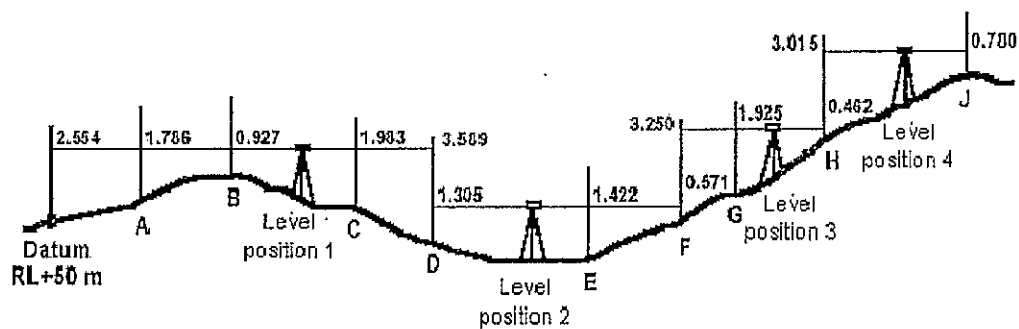
1. Annotate any Five of the following with proper examples (if it requires): (35)
 - (a) Geoid
 - (b) Ellipsoid or Spheroid
 - (c) Cadastral Map
 - (d) Topographic Map
 - (e) Planimetric Map
 - (f) Locator Map
 - (g) Central Meridian
2. (a) What do you understand by map? Write down the importance and function of maps from an Urban Planning perspective. (10)
(b) Mention the classification maps based on general context, scale and subject matter with examples. (10)
(c) What do you understand by thematic maps? Classify thematic maps and provide a brief description of each with necessary figures. (15)
3. (a) A map has a scale of 1:50,000. The area of a park on the map is 6 sq.cm. What is the real area of the park in hectares? (15)
(1 hectare=10,000 sq.m)
(b) Draw a map of an imaginary place with all the necessary elements that a good map layout should have. (10)
(c) What is Optical Centre and Geometric Centre? How can visual balance be obtained in a map layout. (10)
4. (a) Explain your understanding about Datum with necessary diagrams. (10)
(b) Explain the four principal methodological categories of map projection with proper figure and example. (15)
(c) Write about Universal Transverse Mercator (UTM) with respect to Bangladesh. Why is BTM projection system used in Bangladesh instead of having global projection system (UTM)? (10)

Section – B

5. (a) Where chain surveying is recommended and when it is unsuitable? Explain with necessary example. (10)
- (b) Explain the Bowditch's graphical method of traverse adjustment with necessary diagrams. (10)
- (c) A traverse ABCDA is made in the form of a square taking in clockwise order. If the bearing of AB is $120^{\circ}30'$, find the bearing of the other sides. (15)
6. (a) What are the points to be kept in mind while selecting survey stations during chain survey? (06)
- (b) Define the following with necessary sketch and example: (12)
 - i) Whole circle bearing and reduced bearing
 - ii) Fore bearing and back bearing
 - iii) Magnetic declination
- (c) What are the errors that may occur in plane tabling? (10)
- (d) What is index sketch? Draw an index sketch showing the principle of chain survey. (07)
7. (a) What are differences between simple leveling and fly leveling? (05)
- (b) Determine the slope percentage of star marked from the following figure. The distance of the point from 150 contour to 250 contour is 3 inch. The verbal scale of the map is 1 inch=1000ft. (10)



- (c) The following figure is the outcome of leveling survey. Calculate the elevations using collimation and rise-and-falls method and check your result. (20)



8. (a) What are the uses of levelling survey? (05)
- (b) Describe the procedure of developing a contour map through grid method. (15)
- (c) Define the terms "contour interval" and "horizontal equivalent". (05)
- (d) What are the advantages and disadvantages of total station? (10)